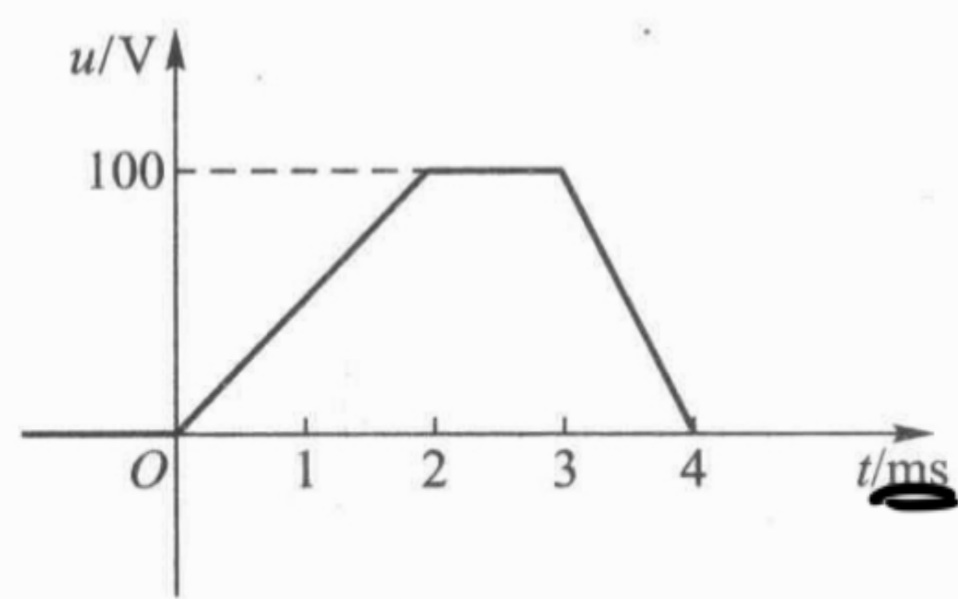


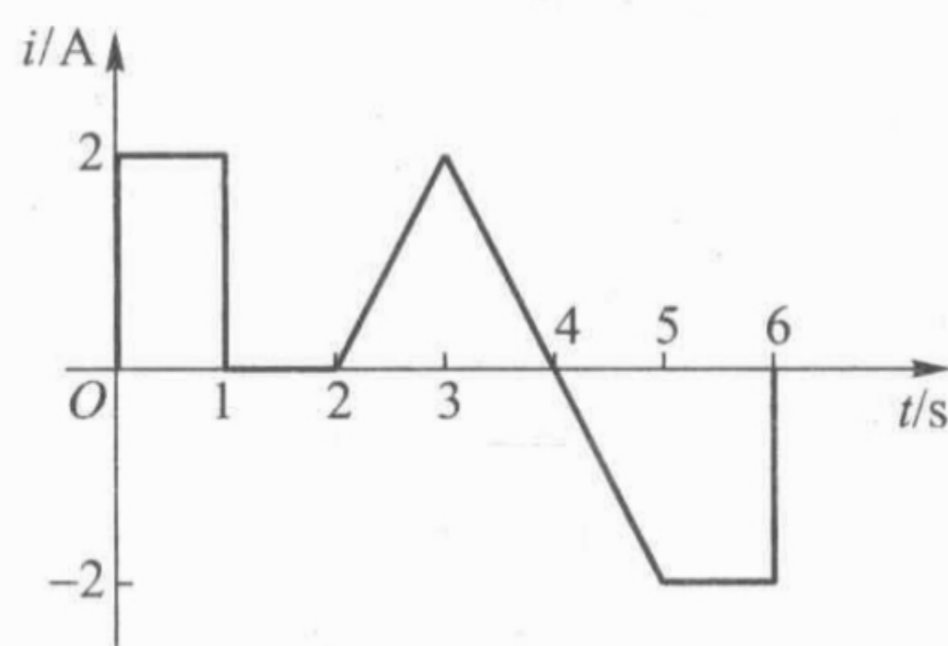
电力作业第5章

5-2 考虑漏电现象,电容器可以用一个电容 C 与漏电阻 R 并联的电路作为模型。若某电容器的模型中 $C=0.1 \mu\text{F}$, $R=150 \text{ k}\Omega$, 外施电压如图题 5-1 所示, 试绘出电容器电流的波形。

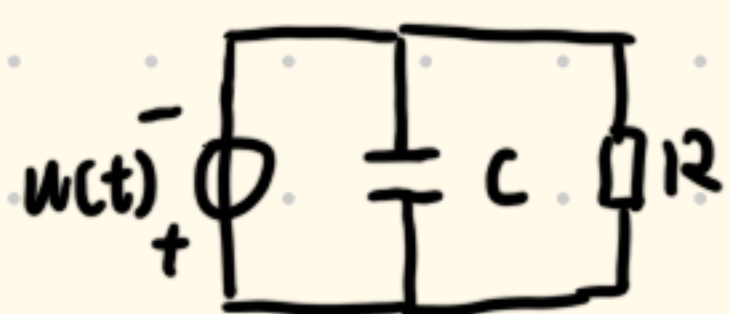
5-3 0.1 F 电容的电流如图题 5-2 所示, 若 $u(0)=0$, 试绘出电容电压的波形。



图题 5-1



图题 5-2



$$i = C \cdot \frac{du}{dt}$$

$$\textcircled{1} \quad 0 \leq t \leq 2 \text{ ms}$$

$$i_c^{(t)} = 0.1 \times 10^{-6} \cdot \frac{100}{2 \times 10^{-3}} = 5 \times 10^{-3} \text{ A} = 5 \text{ mA}$$

$$i_R(t) = \frac{50t}{150} \times 10^{-3} = \frac{t}{3} \text{ mA}$$

$$u(t) = \begin{cases} 50t \text{ V}, & 0 \leq t \leq 2 \text{ ms} \\ 100 \text{ V}, & 2 \text{ ms} < t \leq 3 \text{ ms} \\ 100(t-4) \text{ V}, & 3 \text{ ms} < t \leq 4 \text{ ms} \\ 0, & t > 4 \text{ ms} \end{cases}$$

$$\textcircled{2} \quad 2 \text{ ms} < t \leq 3 \text{ ms}$$

$$i_c^{(t)} = 0$$

$$i_R(t) = \frac{2}{3} \text{ mA}$$

$$\textcircled{3} \quad 3 \text{ ms} < t \leq 4 \text{ ms}$$

$$i_c^{(t)} = 0.1 \times 10^{-6} \cdot \frac{100}{10^{-3}} = 0.01 \text{ A} = 10 \text{ mA}$$

$$i_R(t) = -\frac{2}{3}(t-4) \text{ mA}$$

$$\textcircled{4} \quad t > 4 \text{ ms}$$

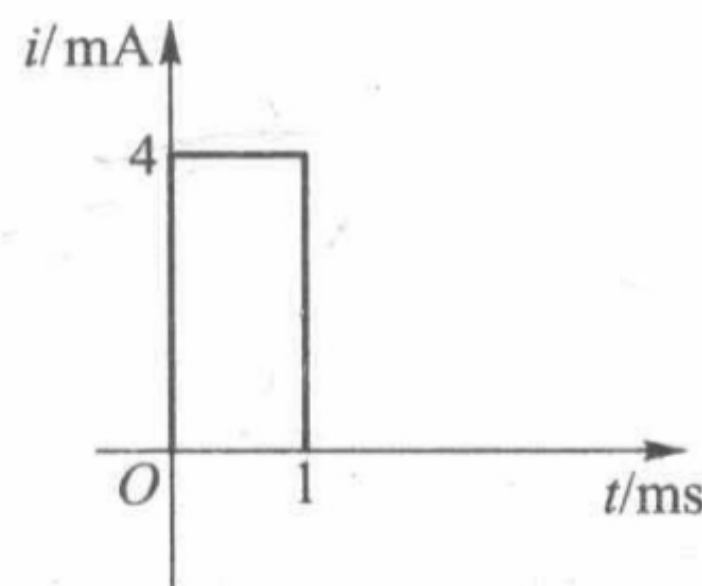
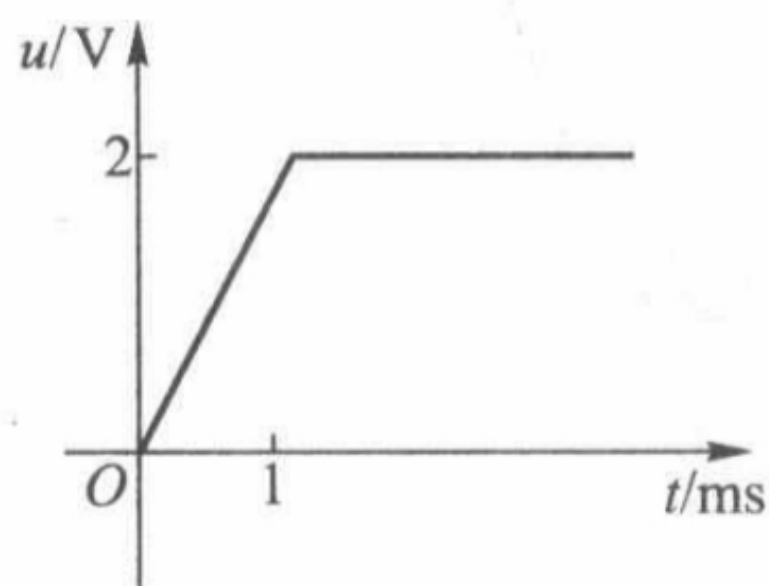
$$i_c^{(t)} = 0$$

$$i_R(t) = 0 \text{ mA}$$

$$i(t) = i_c(t) + i_R(t)$$

$$\Rightarrow i(t) = \begin{cases} 5 + \frac{t}{3} \text{ mA}, & 0 \leq t \leq 2 \text{ ms} \\ \frac{2}{3} \text{ mA}, & 2 \text{ ms} < t \leq 3 \text{ ms} \\ -\frac{2}{3}t + \frac{22}{3}, & 3 \text{ ms} < t \leq 4 \text{ ms} \\ 0, & t > 4 \text{ ms} \end{cases}$$

5-5 图题 5-4 所示为一电容的电压和电流波形。



图题 5-4

(1) 求 C ;

(2) 计算电容在 $0 < t < 1 \text{ ms}$ 期间所得到的电荷;

(3) 计算在 $t = 2 \text{ ms}$ 时吸收的功率;

(4) 计算在 $t = 2 \text{ ms}$ 时储存的能量。

$$\textcircled{1} \quad i = C \cdot \frac{du}{dt} \Rightarrow 4 \times 10^{-3} = C \cdot \frac{2}{1 \times 10^{-3}} \Rightarrow C = 2 \times 10^{-6} \text{ F} = \underline{\underline{2 \mu\text{F}}}$$

$$(2) \quad q = \int_0^1 i \cdot dt = 4 \times 10^{-6} \text{ C}$$

$$(3) \quad p = u \cdot i = 0 \text{ W}$$

$$(4) \quad w = \frac{1}{2} C u^2 = \frac{1}{2} \times 2 \times 10^{-6} \times 4 = 4 \times 10^{-6} \text{ J}$$

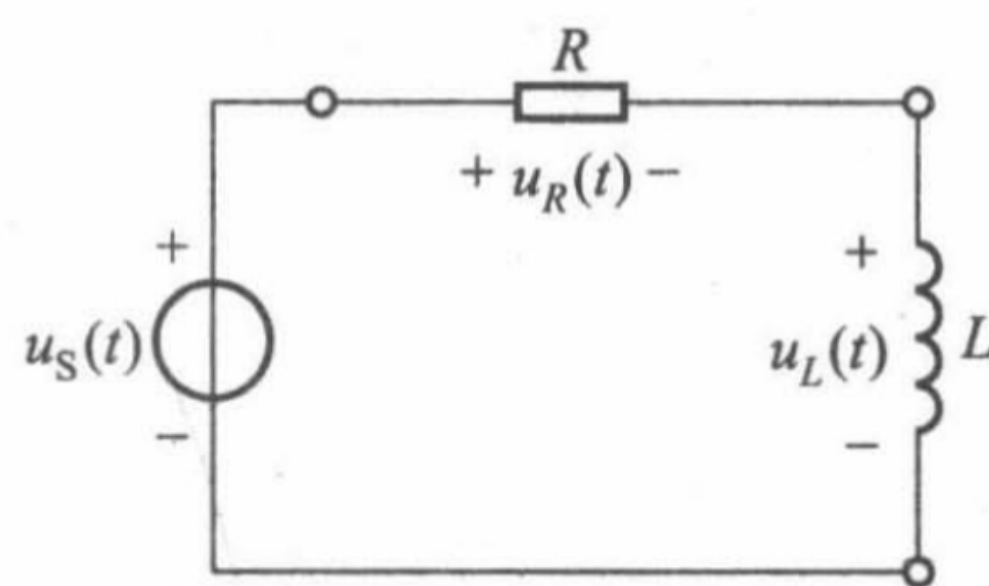
5-7 在图题 5-6 所示电路中 $R = 1 \text{ k}\Omega$, $L = 100 \text{ mH}$, 若

$$u_R(t) = \begin{cases} 15(1 - e^{-10^4 t}) & t > 0 \\ 0 & t < 0 \end{cases}$$

其中 u_R 单位为 V, t 单位为 s。

(1) 求 $u_L(t)$, 并绘波形图;

(2) 求电源电压 $u_S(t)$ 。



图题 5-6

$$(1) \quad u_L(t) = L \frac{di}{dt}$$

① $t < 0$ 时.

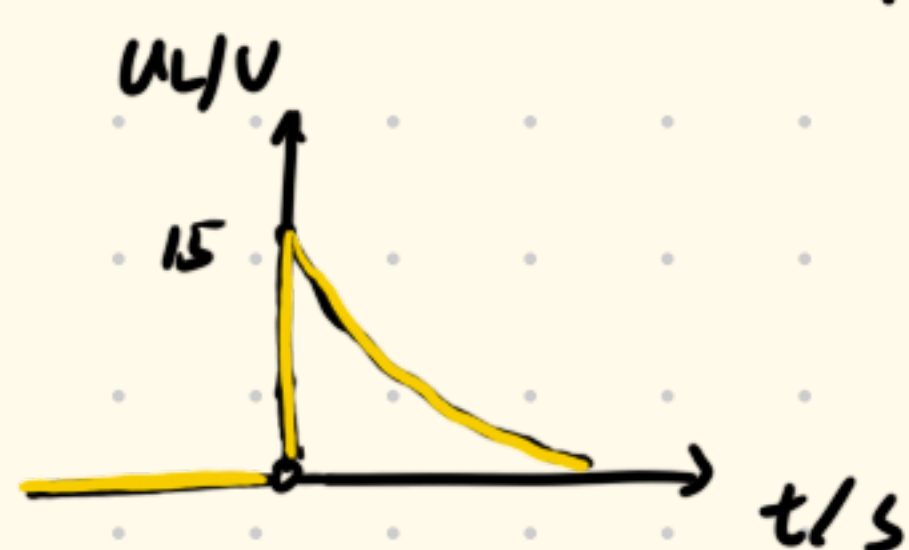
$$i_L = i_R = \frac{u_R(t)}{R} = 0$$

$$\therefore u_L(t) = 0$$

② $t > 0$ 时.

$$i_L = i_R = \frac{u_R(t)}{R} = \frac{15(1 - e^{-10^4 t})}{1 \times 10^3} = 1.5 \times 10^{-2} (1 - e^{-10^4 t}) \text{ A}$$

$$u_L(t) = L \frac{di}{dt} = 100 \times 10^{-3} \times 10^4 \times 1.5 \times 10^{-2} \cdot e^{-10^4 t} = 15 e^{-10^4 t} \text{ V}$$

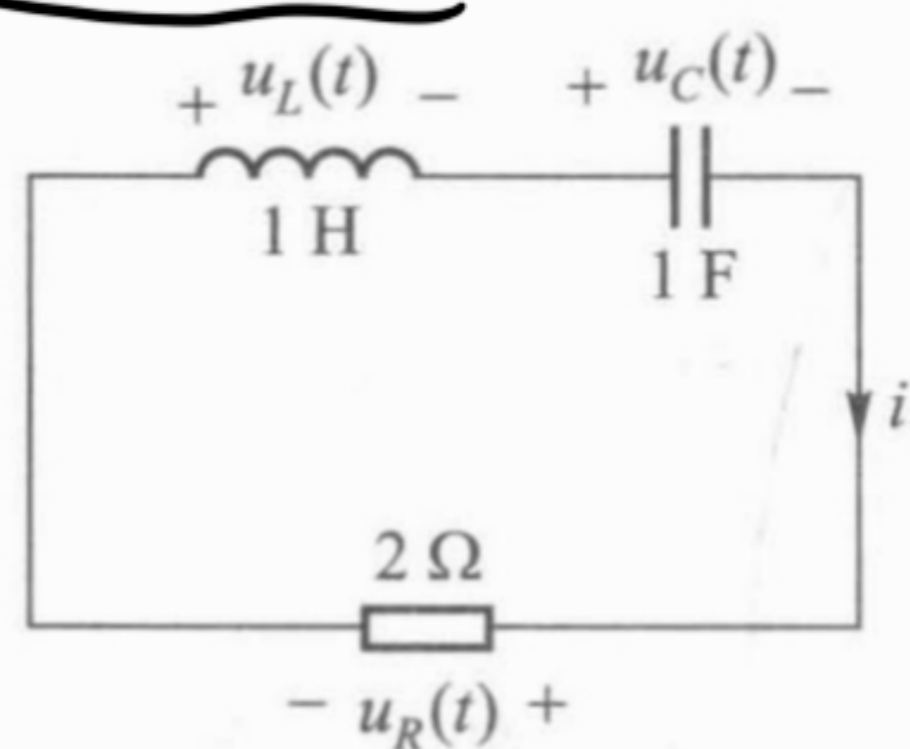


$$u_L(t) = \begin{cases} 15 e^{-10^4 t} \text{ V}, & t > 0 \\ 0, & t < 0 \end{cases}$$

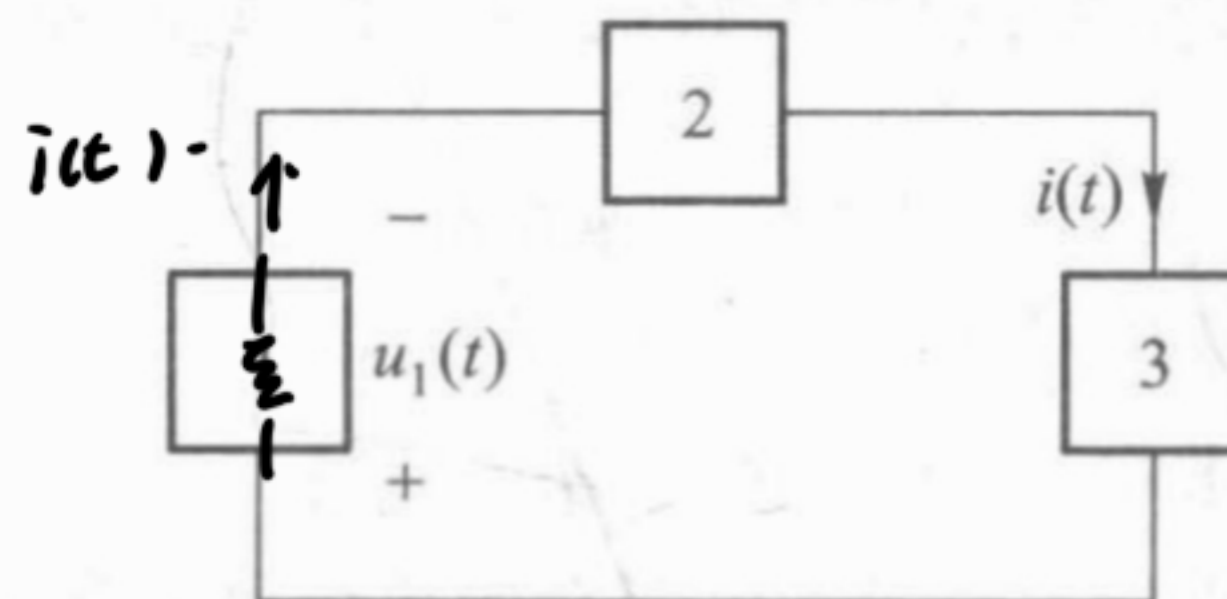
$$(2) \quad u_S(t) = u_L(t) + u_R(t)$$

$$u_S(t) = \begin{cases} 15 \text{ V}, & t > 0 \\ 0, & t < 0 \end{cases}$$

5-12 已知图题 5-11 所示电路由一个电阻 R , 一个电感 L 和一个电容 C 组成。已知 $i(t) = (10e^{-t} - 20e^{-2t}) \text{ A}$, $t \geq 0$; $u_1(t) = (-5e^{-t} + 20e^{-2t}) \text{ V}$, $t \geq 0$ 。若在 $t = 0$ 时, 电路的总储能为 25 J, 试求 R, L, C 之值。



图题 5-10



图题 5-11

$$\frac{di(t)}{dt} = -10e^{-t} + 40e^{-2t} = 2u(t) \Rightarrow \text{1为电感, } L = 0.5 \text{ H}$$

$$t=0. \text{ m.s.} \quad E=25 \text{ J} \quad i=-10 \text{ A.}$$

$$\text{z.B. } E = \frac{1}{2} L i^2 + \frac{1}{2} C u^2.$$

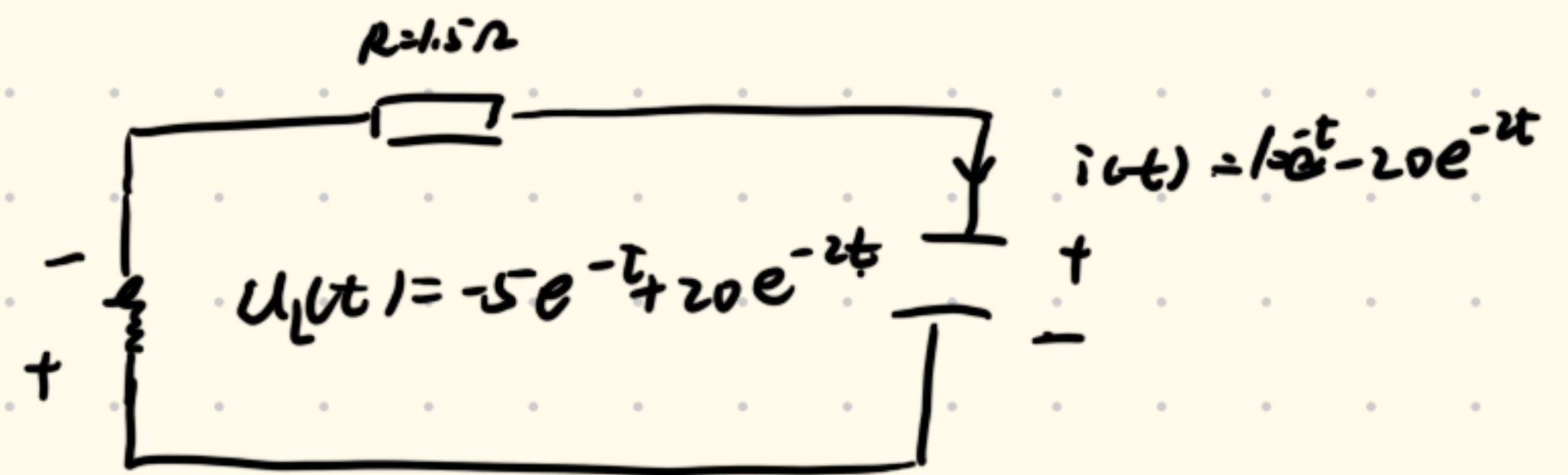
$$\therefore 25 = \frac{1}{2} \times \frac{1}{2} \times 100 + \frac{1}{2} \cdot C u^2.$$

$$\therefore u_C = 0$$

$$\text{z.B. } i(0) = -10 \text{ A.} \quad u_L(0) = 15 \text{ V.}$$

$\therefore$ KVL 方程

$$10 R - 15 = 0 \Rightarrow R = \underline{\underline{1.5 \Omega.}}$$



$$\text{KVL: } i(t) \cdot R + u_C(t) + u_L(t) = 0$$

$$\Rightarrow u_C(t) = 5e^{-t} - 20e^{-2t} - u_L(t) = 5e^{-t} - 20e^{-2t} - (-5e^{-t} + 20e^{-2t}) = 10e^{-t} - 40e^{-2t}.$$

$$i(t) = C \cdot \frac{du}{dt} \Rightarrow C = \frac{du}{\frac{di}{dt} \cdot i(t)} = \frac{10e^{-t} - 40e^{-2t}}{10e^{-t} - 20e^{-2t}} = \underline{\underline{1 \text{ F}}}$$